University of Zurich[UZH]

Macro Forecasting with the TFT and Fed Speeches

Minna Heim, Chris Traill and Anna Zeitz

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- Macro forecasts matter for
  - Governments
  - Financial markets
  - Central banks

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  - **Central banks**
- The Federal Reserve (Fed) has a **dual mandate**:
  1. Price stability
  2. Maximum employment
- Accurate forecasts of inflation, unemployment, and GDP are crucial for interest rate decisions

Can we do better than standard Macro Models?

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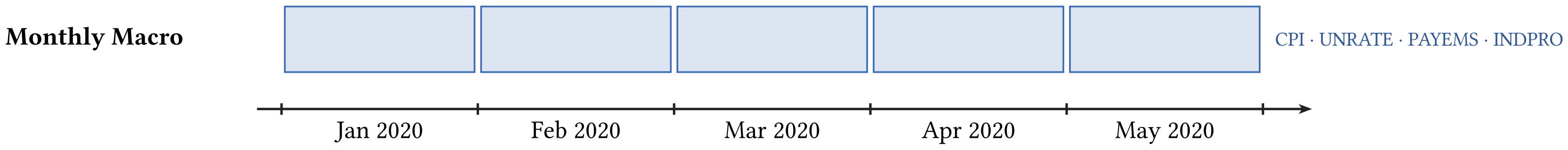
⇒ **Do Fed speeches contain information useful to forecast macroeconomic indicators?**

Data Alignment



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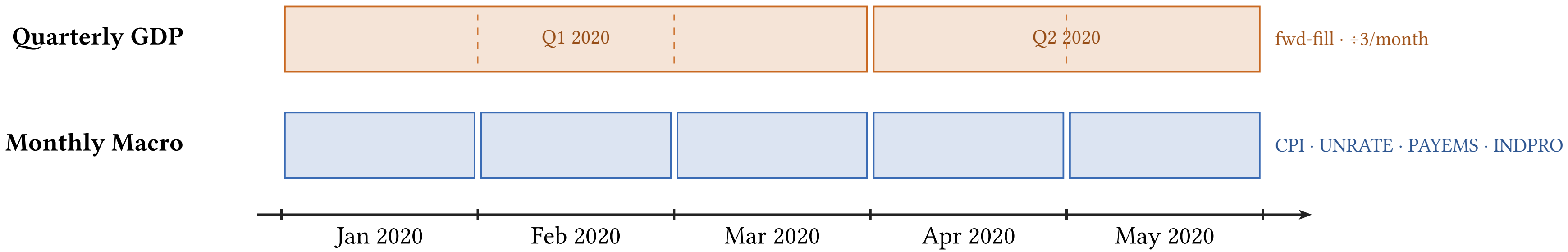

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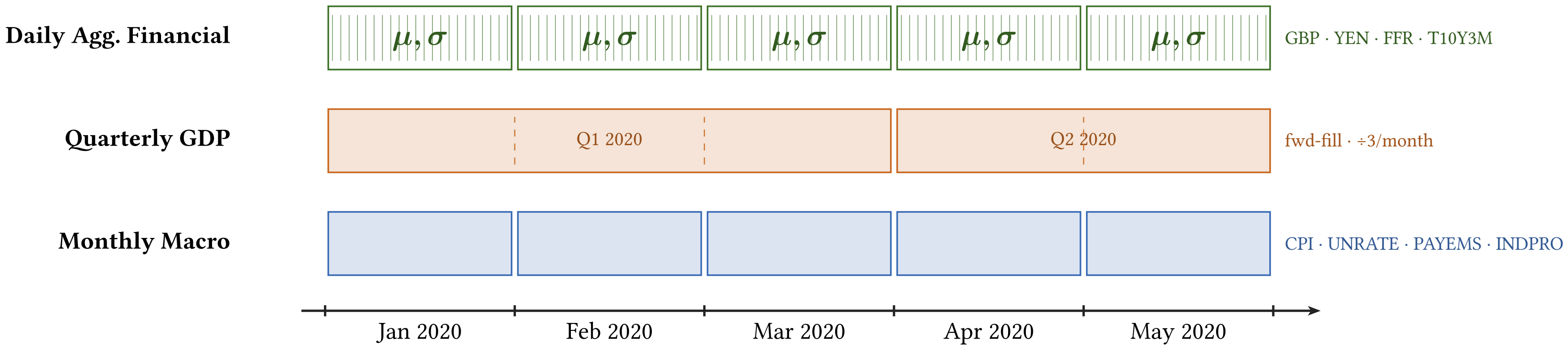

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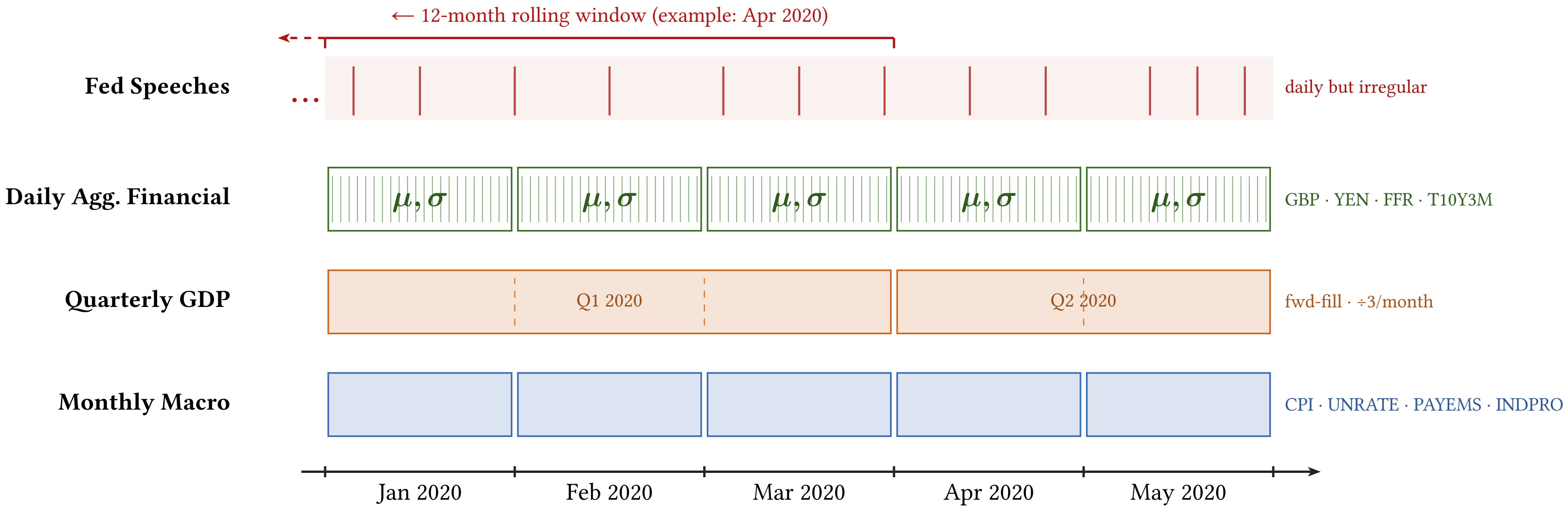

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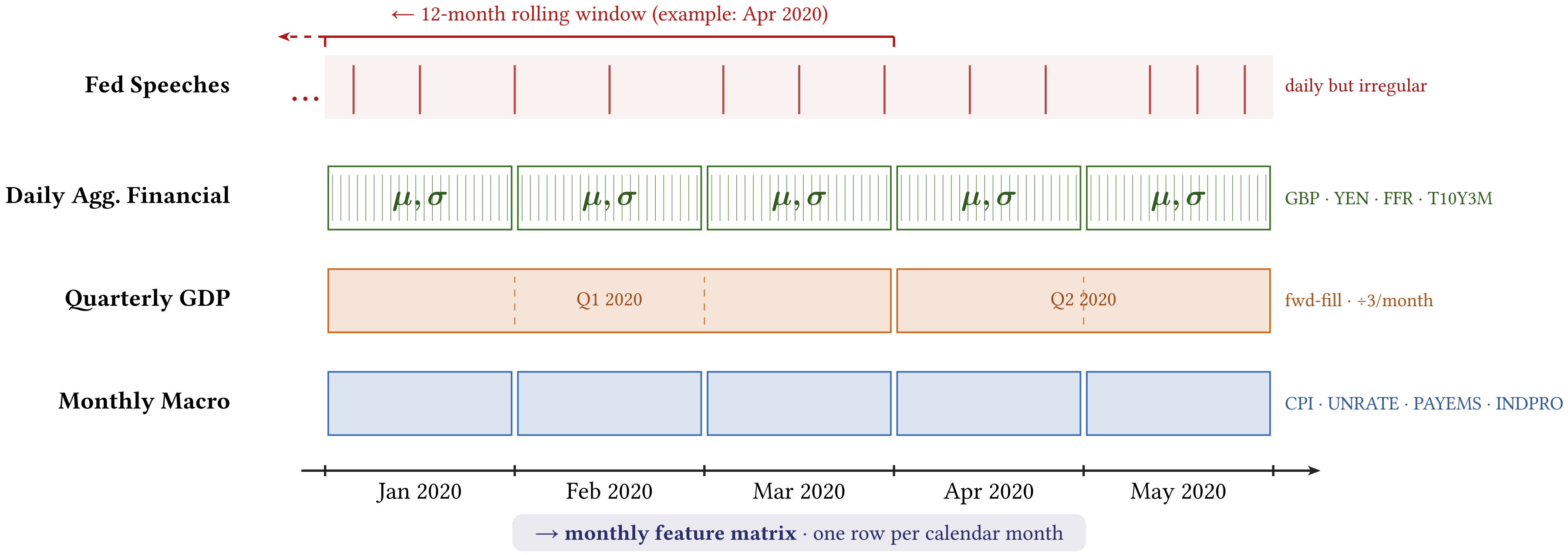

Figure 1: Data alignment pipeline.

Speeches are taken from Campiglio et al. (2023) over period 1986 — 2023.

Models

Benchmarks

AR(p)

$$y_t = c + \sum_{i=1}^{p} \varphi_i y_{t-i} + \varepsilon_t$$

ARIMA(p, d, q)

$$\Delta^d y_t = c + \sum_{i=1}^{p} \varphi_i \Delta^d y_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t$$

Integration: d-th difference of y_t

MA component: q lags of residuals ε_t

Temporal Fusion Transformer

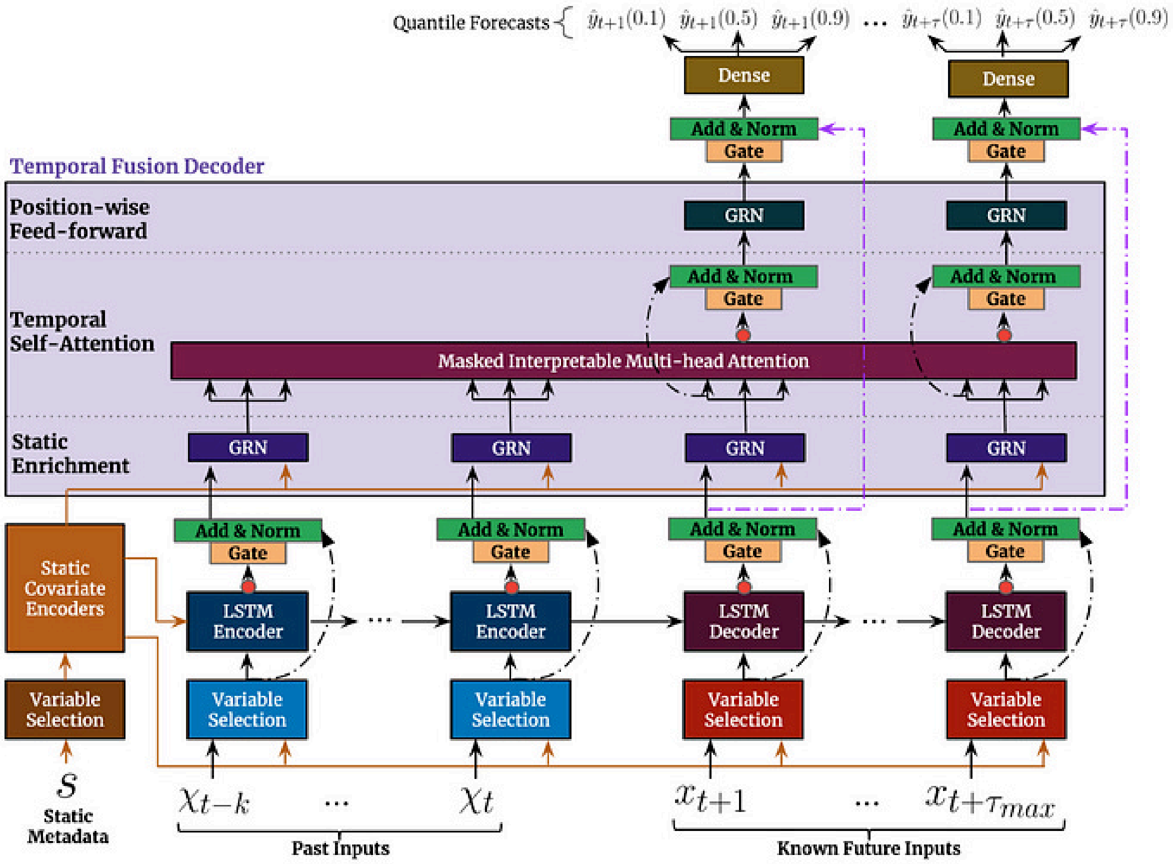

Figure 2: Model Architecture of the TFT (Lim et al., 2021)

Speech Embeddings

Speech Embeddings

Two models chosen for complementary perspectives:

1. **FinBERT**: BERT-based, pre-trained on financial statements (Yang et al., 2020)
2. **FOMC-RoBERTa**: fine-tuned on FOMC data, hawkish/dovish classification (Shah et al., 2023)

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Processing: chunk-and-average (512-token windows, 128-token overlap)

- Assumption: information spread across entire speech, not just the beginning

Dimensionality reduction: $768 \Rightarrow X$ dims via Principal Component Analysis or Factor Analysis

- X is treated as hyperparameter

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3. **Attention-based**: key = embedding (after dimensionality reduction), query = mean macro state
   - Given current macro conditions, which past speeches are most informative?
   - Fitted on training data only $\Rightarrow$ no leakage!

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All methods also track: voter speech ratio, chair speech ratio, gender share

Hyperparameter Tuning

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1. **Stage 1: Architecture (Macro-Only TFT)**
    - Bayesian search via Optuna (TPE sampler)
    - 2-fold CV, up to 50 trials
    - Includes: encoder length, hidden size, normalizer

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1. **Stage 1: Architecture (Macro-Only TFT)**
    - Bayesian search via Optuna (TPE sampler)
    - 2-fold CV, up to 50 trials
    - Includes: encoder length, hidden size, normalizer
2. **Stage 2: Speech Embedding Params**
    - Architecture fixed from Stage 1
    - Tune: aggregation, reduction, PCA dims, speech window

Evaluation Protocol

- **Walk-forward cross-validation**: expanding window, never use future data

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  - Context: training data + model's own previous predictions
  - Same assumption for AR, ARIMA and TFT

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$\Rightarrow$ All models evaluated on the same information set

Results

Holdout Evaluation

Holdout Metrics: All Models & Horizons

		CPI		GDP		UNRATE	
h	**Model**	**MAE ↓**	**RMSE ↓**	**MAE ↓**	**RMSE ↓**	**MAE ↓**	**RMSE ↓**
3	AR	0.00218	0.00252	0.005	0.00511	0.24238	0.3012
	ARIMA	**0.00122**	**0.00146**	0.005	0.00511	0.25415	0.31466
	TFT	0.0021	0.00239	0.00297	0.00318	**0.17877**	**0.20819**
	TFT+Emb	0.00239	0.00269	**0.00279**	**0.00302**	0.25455	0.32353
6	AR	0.00218	0.00252	0.005	0.00511	**0.24238**	**0.3012**
	ARIMA	**0.00122**	**0.00146**	0.005	0.00511	0.25415	0.31466
	TFT	0.00359	0.0038	0.00322	0.00342	0.71827	0.77017
	TFT+Emb	0.00269	0.00302	**0.00306**	**0.00328**	0.63271	0.64908
12	AR	0.00218	0.00252	0.005	0.00511	**0.24238**	**0.3012**
	ARIMA	**0.00122**	**0.00146**	0.005	0.00511	0.25415	0.31466
	TFT	0.00311	0.00343	0.00342	0.00359	2.23273	2.29207
	TFT+Emb	0.00342	0.00367	**0.00102**	**0.00119**	1.76382	1.98447

Table 1: Main Results. Bold = lowest MAE and RMSE per (horizon, target).

Answering Our Research Question

	CPI			GDP			UNRATE		
h	$\text{RMSE}_{\text{TFT}} \downarrow$	$\text{RMSE}_{\text{+Emb}} \downarrow$	relRMSE $\downarrow$	$\text{RMSE}_{\text{TFT}} \downarrow$	$\text{RMSE}_{\text{+Emb}} \downarrow$	relRMSE $\downarrow$	$\text{RMSE}_{\text{TFT}} \downarrow$	$\text{RMSE}_{\text{+Emb}} \downarrow$	relRMSE $\downarrow$
3	0.00239	0.00269	1.1255	0.00318	0.00302	**0.9497**	0.20819	0.32353	1.5540
6	0.0038	0.00302	**0.7947**	0.00342	0.00328	**0.9591**	0.77017	0.64908	**0.8428**
12	0.00343	0.00367	1.0700	0.00359	0.00119	**0.3315**	2.29207	1.98447	**0.8658**

Table 2: Relative RMSE. Bold if TFT with Embeddings better than TFT, per (horizon, target).

Holdout Predictions (h=12)

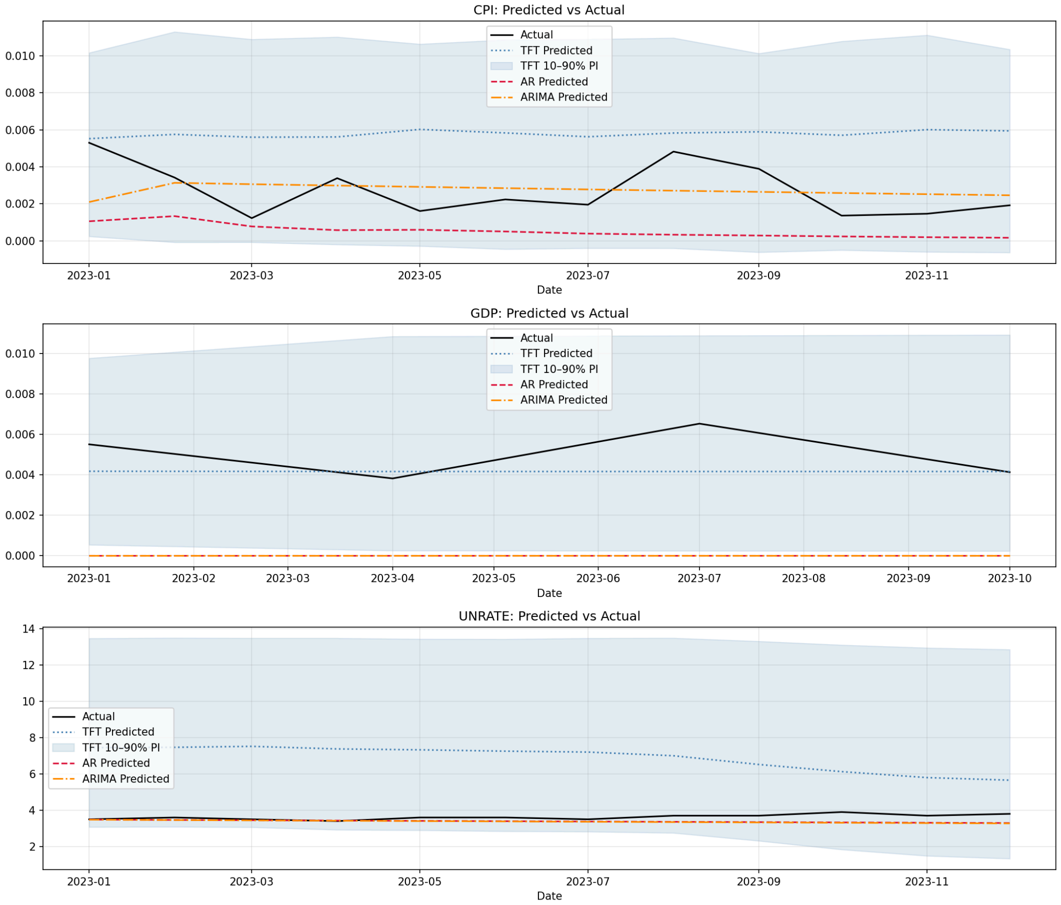

Figure 3: Holdout Predictions h=12, Macro Only

Holdout Predictions (h=12)

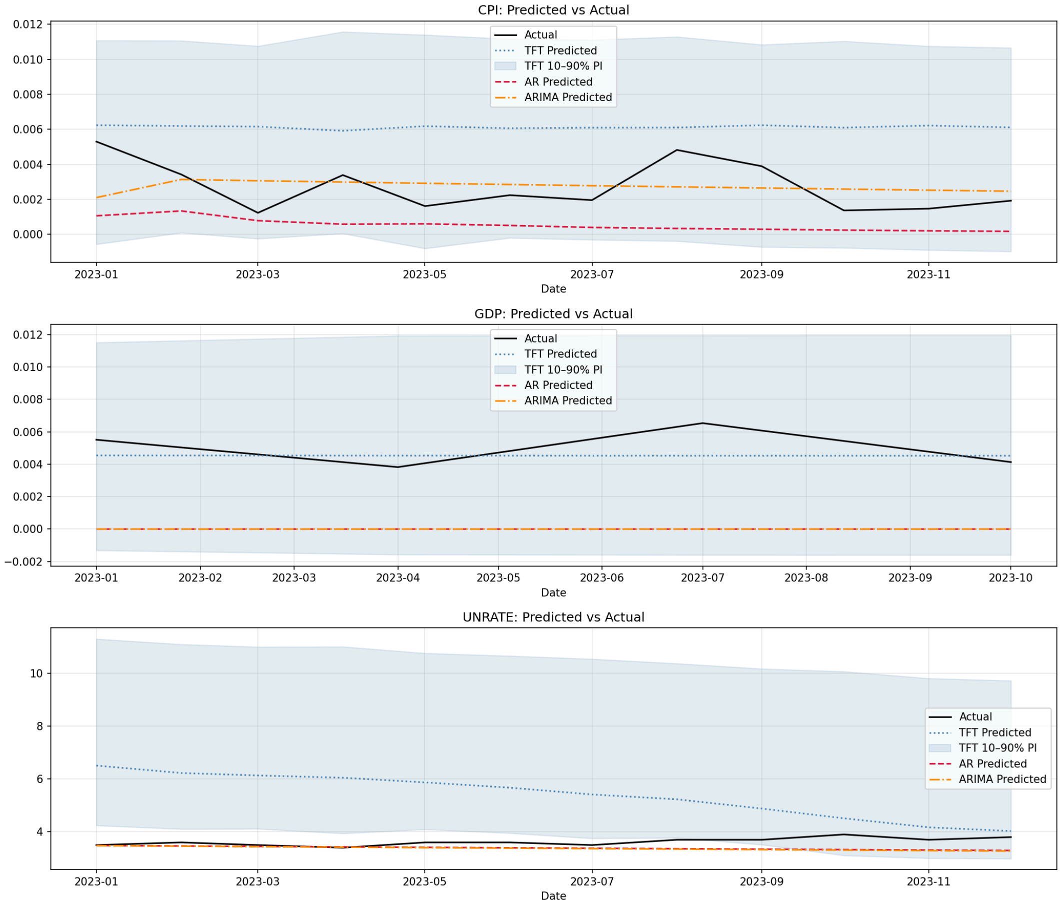

Figure 4: Holdout Predictions h=12, Including Embeddings

Robustness Checks

- Using unrelated text (German Texts of Franz Kafka)
- Use only first 512 tokens of speech

⇒ **Mixed results**: e.g., better than TFT + Emb. at CPI, $h = 3$ but worse at CPI, $h = 6$ ⇒ More details: Table 9 in Appendix

Future Work

- Hyperparameter tuning: switch to one stage tuning!
  - Context-dependent attention
  - Only FOMC speeches
- Explore New Methods:
  - Try out SSMs for forecasting
  - Combine with TFT: replace LSTM with SSM for sequence encoder
- Add vintages: use real-time macro data (as available at forecast time) for more realistic evaluation
- Extend dataset: Working Paper (Lustenberger et al., 2026)
  - 10,000+ speeches since 1914, continuously updated

Conclusion

- AR and ARIMA dominate CPI and UNRATE
- TFT great for GDP forecasts
- TFT naturally includes uncertainty bounds

⇒ **Speeches improve TFT forecasts at medium horizon length**

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- AR and ARIMA dominate CPI and UNRATE
- TFT great for GDP forecasts
- TFT naturally includes uncertainty bounds

⇒ **Speeches improve TFT forecasts at medium horizon length**

- But: not necessarily due to speech content but due to more stable TFT

⇒ **Conclusion: we struggle to beat statistical benchmarks**

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Appendix

Variable List: Macroeconomic Variables

1. **Target variables**
   - CPI, UNRATE, GDP (quarterly: forward-fill missing months) $\Rightarrow$ AR and ARIMA only take targets
2. **Covariates**
   - Monthly variables: payroll (PAYEMS), industrial production (INDPRO), hours worked in manufacturing (AWHMAN), OECD composite leading indicator (USACLI)
   - Weekly / daily variables: exchange rates (GBP, YEN), effective federal funds rate (FFR), Fed balance sheet (WALCL)
     - Higher-frequency variables: take mean **and** std per month (volatility)
   - Lags at 1, 2, 6, 12 months for all target variables
3. **Preprocessing**: log-difference of CPI, GDP, PAYEMS, INDPRO, AWHMAN

Data Frame:

date	CPI	PAYEMS	INDPRO	UNRATE	AWHMAN	USACLI	GDP	GBP_mean	GBP_std	YEN_mean	YEN_std	FFR_mean
1986-01	0.0036	0.0012	0.0055	6.7	-0.0049	0.0002	0.0048	1.4244	0.0227	199.8905	3.6638	8.1448
1986-02	-0.0018	0.0012	-0.007	7.2	0.0	0.0001	0.0048	1.4297	0.0352	184.8516	4.586	7.86
1986-03	-0.0055	0.0009	-0.0071	7.2	0.0	-0.0001	0.0048	1.4674	0.0154	178.6938	1.8252	7.479
1986-04	-0.0037	0.0019	0.0014	7.1	-0.0049	-0.0001	0.0028	1.4985	0.0363	175.0918	5.1994	6.988
1986-05	0.0028	0.0013	0.0015	7.2	0.0049	-0.0002	0.0028	1.5211	0.021	167.0314	3.4488	6.8529
1986-06	0.0037	-0.0009	-0.004	7.2	0.0	-0.0003	0.0028	1.5085	0.0157	167.5419	2.5161	6.9167

FFR_std	T10Y2Y_mean	T10Y2Y_std	dissent_rate_d	dissent_net_hawk	dissent_net_hawk_s	n_tighter_sum	n_easier_sum	any_dissent_recen	days_since_fomc	days_to_fomc	fomc_cycle_pos	meeting_this_mon
1.0103	1.0514	0.0368	0.0922	0.25	2.0	5.0	3.0	1.0	15.0	42.0	0.2632	0.0
0.1221	0.7368	0.2171	0.0922	0.25	2.0	5.0	3.0	1.0	46.0	11.0	0.807	1.0
0.2745	0.564	0.0644	0.1131	0.0	0.0	5.0	5.0	1.0	17.0	31.0	0.3542	0.0
0.2155	0.6041	0.0734	0.1292	0.0	0.0	5.0	5.0	1.0	0.0	49.0	0.0	1.0
0.085	0.6424	0.0375	0.1131	0.0	0.0	5.0	5.0	1.0	30.0	19.0	0.6122	1.0
0.175	0.6148	0.1007	0.114	0.0	0.0	5.0	5.0	1.0	12.0	38.0	0.24	0.0

Figure 5: Head of the main data frame

Target Vars

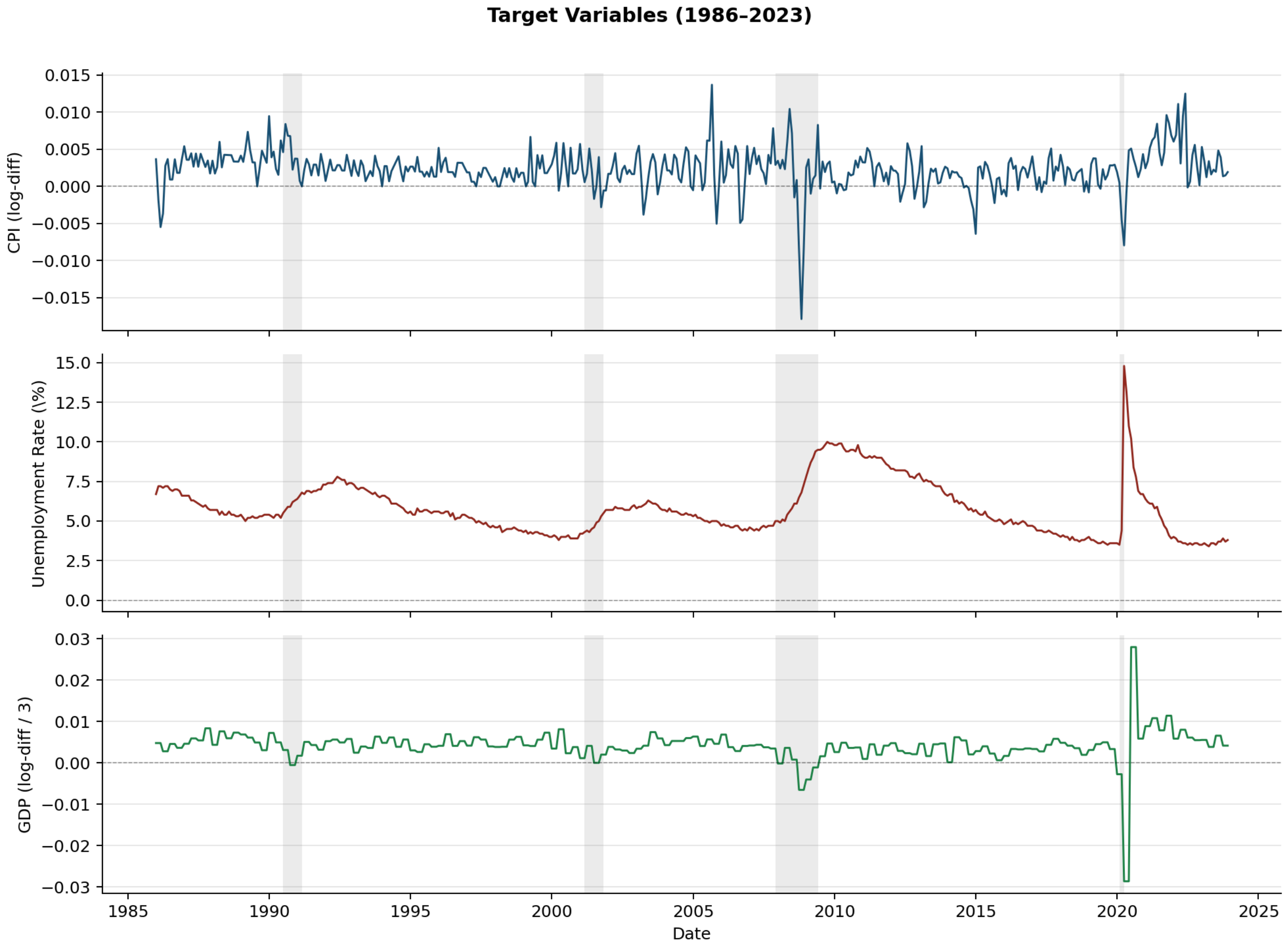

Figure 6: Time series of the target variables

Variable List

The TFT takes four different types of variables:

1. **Static categoricals**: time-invariant, categorical
    - Identifier and frequency of each time series
2. **Static reals**: time-invariant, continuous
    - Number downloads time series from FRED, how long time series has existed
3. **Time-Varying Known Reals**: time-varying, known
    - Day of week, month, year; if day is holiday
    - FOMC Meeting dates
4. **Time-Varying Unknown Reals**: time-varying, unknown
    - Macroeconomic variables, speech embeddings, FOMC dissents

TFT: Input Types

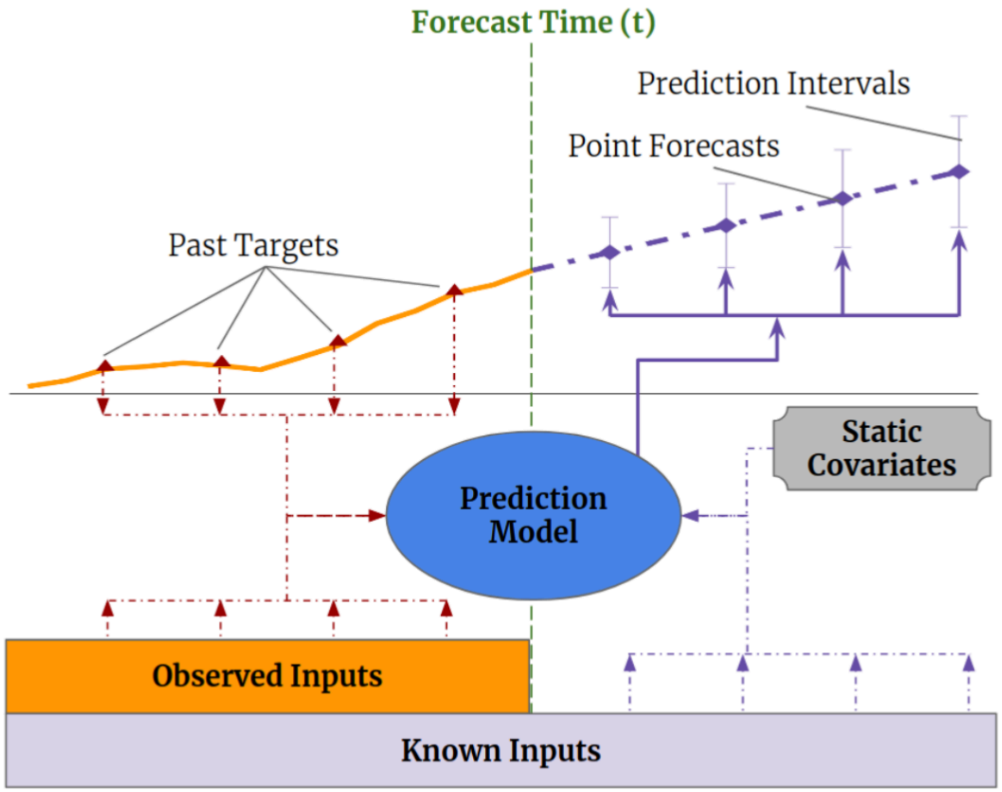

Figure 7: Temporal Fusion Transformer: Input Types

Dimensionality Reduction

FinBERT and FOMC-RoBERTa embeddings are of dimension 768 per speech

$\Rightarrow$ Dimensionality reduction needed for TFT input

We test two methods (number of components X is hyperparameter, fit only on training data):

1. **PCA**: projects embeddings onto X principal components
   - 20 components $\approx$ 80% of variance explained
2. **Factor Analysis (FA)**: models embeddings as linear combination of X latent factors
   - Unlike PCA: explicitly models noise, focuses on **shared** variance
   - Parameters estimated via EM algorithm

Hyperparameter Tuning: Stage 1 Search Space

Architecture tuned on macro-only TFT (per target × horizon):

Parameter	Type	Range / Values	Nr. of Trials
Encoder length	int	12 – 48	20
Hidden size	categorical	{8, 16, 32, 64, 128, 256}	20
Hidden continuous size	categorical	{2, 4, 8, 16}	20
Dropout	float	0.05 – 0.55	20
Learning rate	log-uniform	10^{-4} – 0.15	20
Normalizer	categorical	{encoder-none, group}	20

Table 3: Stage 1: Hyperparameter search space and number of trials.

Fixed: `lstm_layers = 2`, `attention_head_size = 2`,

`max_epochs = 50`, `patience = 15`, `batch_size = 128`

Hyperparameter Tuning: Stage 2 Search Space

Speech embedding params tuned (architecture fixed from Stage 1):

Parameter	Type	Range / Values
Aggregation	categorical	{mean, decay, attention}
Reduction	categorical	{pca, fa}
PCA components (X)	int	5 – 30
Speech window (months)	int	3 – 12

Table 4: Stage 2: Hyperparameter search space and number of trials.

Tuned separately for each of 9 combinations: {CPI, GDP, UNRATE} $\times$ {3, 6, 12} months

Hyperparameter Tuning: Stage 2 – Optimal Hyperparameters

	FOMC-RoBERTa			FinBERT		
Target	**h=3**	**h=6**	**h=12**	**h=3**	**h=6**	**h=12**
CPI	Mean PCA, n = 21 W = 3 MAE = 0.001847	Attention FA, n = 10 W = 8 MAE = 0.001849	**Attention** **PCA, n = 12** **W = 9** **MAE = 0.001842**	**Attention** **FA, n = 20** **W = 5** **MAE = 0.001846**	**Decay** **PCA, n = 23** **W = 3** **MAE = 0.001846**	Mean PCA, n = 8 W = 7 MAE = 0.001847
GDP	**Mean** **PCA, n = 8** **W = 7** **MAE = 0.00154**	Attention FA, n = 12 W = 8 MAE = 0.00171	Attention FA, n = 5 W = 10 MAE = 0.00152	Attention FA, n = 6 W = 12 MAE = 0.00170	**Attention** **FA, n = 10** **W = 8** **MAE = 0.00157**	**Decay** **PCA, n = 20** **W = 12** **MAE = 0.00151**
UNRATE	Decay FA, n = 12 W = 11 MAE = 1.31293	Attention FA, n = 6 W = 12 MAE = 1.31501	**Mean** **PCA, n = 8** **W = 7** **MAE = 1.11597**	**Mean** **FA, n = 23** **W = 5** **MAE = 1.26152**	**Mean** **PCA, n = 14** **W = 4** **MAE = 1.30896**	Mean PCA, n = 25 W = 4 MAE = 1.20808

Table 5: Stage 2: Optimal hyperparameters per (target, horizon).

where W refers to the speech window size and n to the optimal number of components for the dimensionality reduction. The best result per target×horizon is shown in bold, and h stands for forecast horizon.

Holdout Predictions (h=6)

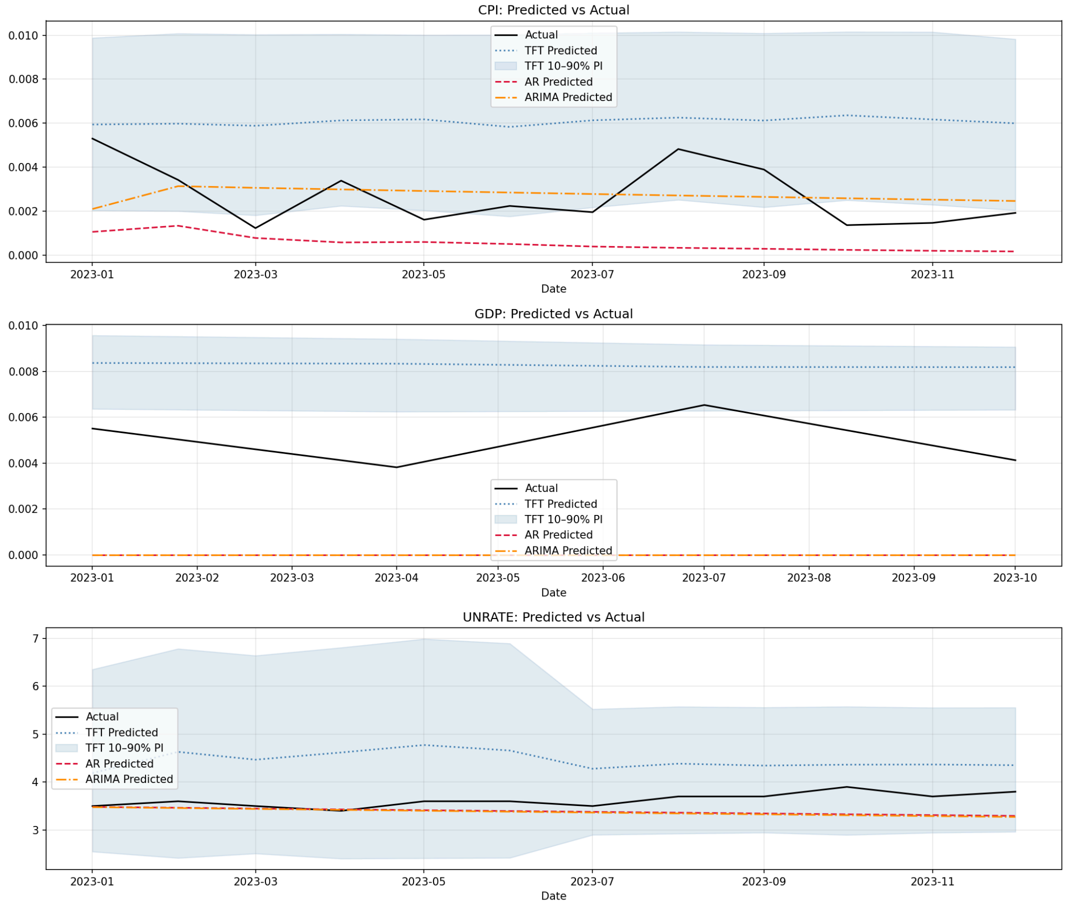

Figure 8: Holdout Predictions h=6, Macro Only

Holdout Predictions (h=6)

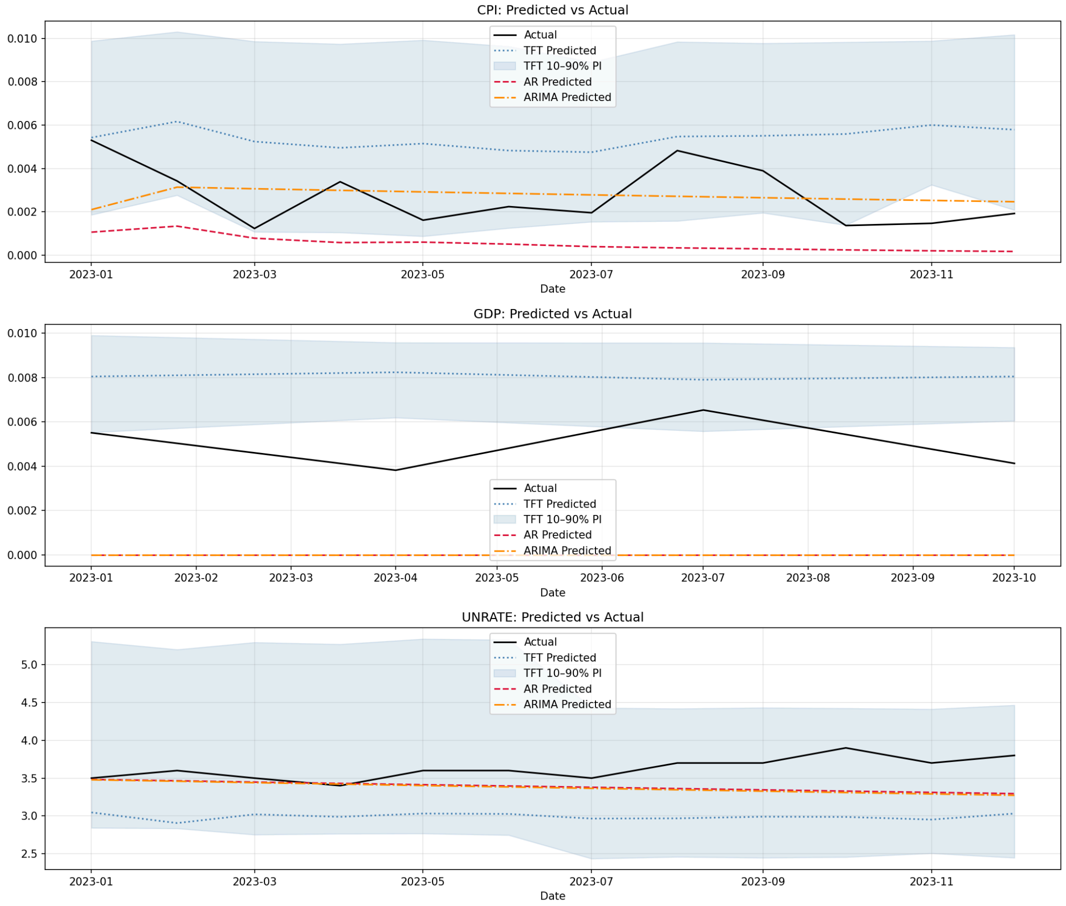

Figure 9: Holdout Predictions h=6, Including Embeddings

Holdout Predictions (h=3)

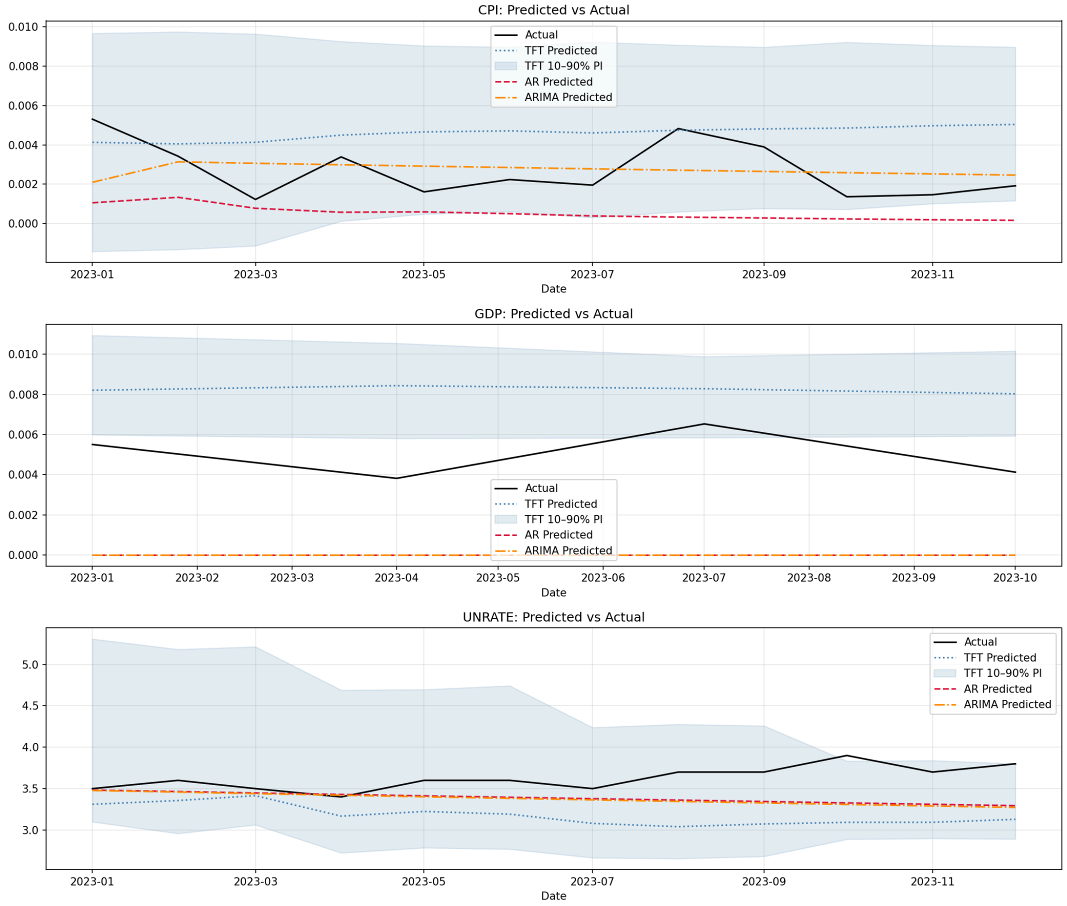

Figure 10: Holdout Predictions h=3, Macro Only

Holdout Predictions (h=3)

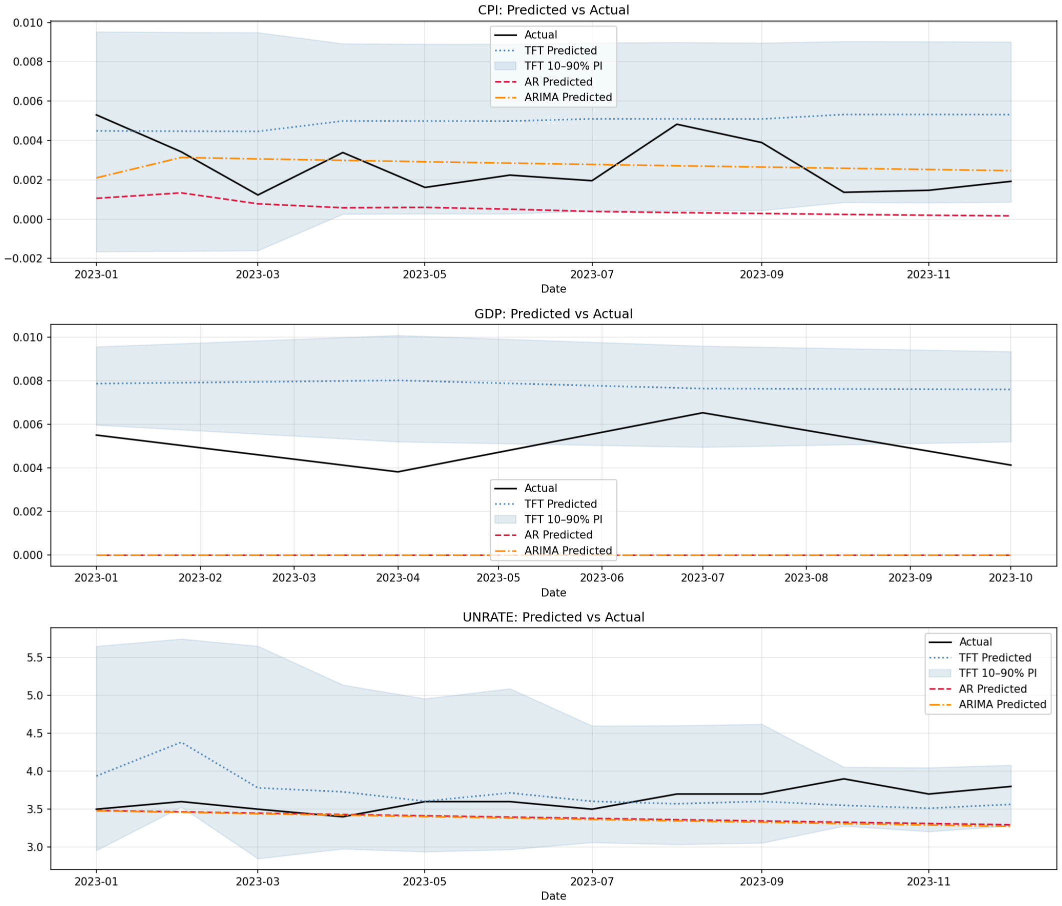

Figure 11: Holdout Predictions h=3, Including Embeddings

Variable Selection

Variable	CPI	GDP	UNRATE	Mean
GDP	0.0191	0.0202	0.8933	**0.3109**
CPI_lag_1	0.2202	0.0126	0.0012	**0.078**
YEN_std	0.0256	0.1005	0.0009	**0.0423**
GDP_lag_2	0.0274	0.0759	0.0013	**0.0349**
week_of_year	0.0631	0.0171	0.0006	**0.027**
GDP_lag_12	0.0539	0.0224	0.0011	**0.0258**
dissent_net_hawk_sum	0.0174	0.0135	0.0312	**0.0207**
PAYEMS_lag_2	0.036	0.017	0.0011	**0.0181**
GDP_lag_1	0.005	0.048	0.0014	**0.0181**
AWHMAN	0.0294	0.0212	0.0014	**0.0174**

Table 6: Top 10 encoder variables, $h = 12$

Variable Selection

Variable	CPI	GDP	UNRATE	Mean
time_idx	0.0265	0.9993	0.0198	**0.3485**
days_to_fomc	0.0116	0	0.5956	**0.2024**
fomc_cycle_pos	0.4511	0.0001	0.0332	**0.1615**
days_since_fomc	0.1966	0.0001	0.0132	**0.07**
meeting_this_month	0.0843	0.0001	0.0297	**0.038**
is_holiday	0.0537	0.0002	0.0258	**0.0266**
week_of_year	0.0469	0.0001	0.0291	**0.0254**
day_of_week	0.0345	0.0001	0.0359	**0.0235**
month	0.0383	0	0.0235	**0.0206**

Table 7: Top 10 decoder variables, $h = 12$

Variable Selection

Variable	CPI	GDP	UNRATE	Mean
INDPRO_meta_popularity	0.0054	0.8715	0.0332	**0.3034**
PAYEMS_meta_popularity	0.2495	0.0012	0.0165	**0.0891**
GDP_meta_years_of_history	0.0773	0.0545	0.0372	**0.0563**
INDPRO_meta_units	0.1438	0.0011	0.0151	**0.0534**
CPI_meta_units	0.0316	0.0007	0.0791	**0.0371**
GDP_meta_units	0.0762	0.0006	0.0222	**0.033**
CPI_meta_popularity	0.072	0.0014	0.0237	**0.0323**
GDP_meta_frequency	0.0477	0.0014	0.0186	**0.0226**
UNRATE_meta_units	0.018	0.0071	0.0408	**0.022**
UNRATE_meta_frequency	0.0097	0.0007	0.049	**0.0198**

Table 8: Top 10 static variables, $h = 12$

Holdout Metrics: Robustness

		CPI		GDP		UNRATE	
h	**Model**	**MAE ↓**	**RMSE ↓**	**MAE ↓**	**RMSE ↓**	**MAE ↓**	**RMSE ↓**
3	AR	0.00218	0.00252	0.005	0.00511	0.24238	0.3012
	ARIMA	**0.00122**	**0.00146**	0.005	0.00511	0.25415	0.31466
	TFT	0.0021	0.00239	0.00297	0.00318	**0.17877**	**0.20819**
	TFT+Emb	0.00239	0.00269	**0.00279**	**0.00302**	0.25455	0.32353
	Kafka	0.00238	0.00268	0.0034	0.0036	0.78076	0.81517
	512 Tokens	0.00233	0.00264	0.0033	0.00348	0.27192	0.29756
6	AR	0.00218	0.00252	0.005	0.00511	**0.24238**	**0.3012**
	ARIMA	**0.00122**	**0.00146**	0.005	0.00511	0.25415	0.31466
	TFT	0.00359	0.0038	0.00322	0.00342	0.71827	0.77017
	TFT+Emb	0.00269	0.00302	**0.00306**	**0.00328**	0.63271	0.64908
	Kafka	0.00349	0.0037	0.00309	0.0033	0.5526	0.59865
	512 Tokens	0.00337	0.00359	0.00325	0.00346	0.4984	0.59655
12	AR	0.00218	0.00252	0.005	0.00511	**0.24238**	**0.3012**
	ARIMA	**0.00122**	**0.00146**	0.005	0.00511	0.25415	0.31466
	TFT	0.00311	0.00343	0.00342	0.00359	2.23273	2.29207
	TFT+Emb	0.00342	0.00367	**0.00102**	**0.00119**	1.76382	1.98447
	Kafka	0.0031	0.00336	0.0011	0.00149	1.53273	1.65122
	512 Tokens	0.00409	0.00435	0.00294	0.00314	1.34398	1.54827

Table 9: Robustness with Kafka and first 512 tokens embeddings. Bold = lowest MAE and RMSE per (horizon, target).